MIL-C-43293A(GL)

4 June 1973

SUPERSEDING
MIL-C-43293(GL)
25 November 1964

MILITARY SPECIFICATION

CASES, FIELD, OFFICE MACHINES, SURVEYING EQUIPMENT, STEEL

1. SCOPE

1.1 Scope. This specification covers steel cases for storage of specific office type machines required to support surveying operations in the field (see 6.1).

1.2 Classification. The surveying equipment cases shall be of the following types, as specified (see 6.2):

Type I - For adding machine.
Type II - For calculating machine.
Type III - For typewriter or computing machine.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester.
QQ-S-630	-	Steel, Bar, Carbon, Hot Rolled, (Merchant Quality).
QQ-S-634	-	Steel, Bar, Carbon, Gold Finished (Standard Quality).
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
ZZ-T-351	-	Tips; Rubber and Synthetic-rubber for Crutches, Furniture, etc.
MMM-A-1617	-	Adhesive, Rubber Base, General Purpose.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

FSC 7420

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MILITARY

MIL-C-496	-	Clips, End, Strap.
MIL-W-530	-	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-B-543	-	Buckles, Tongueless and Web Strap.
MIL-T-704	-	Treatment and Painting of Material.
MIL-L-2045	-	Loops, Strap Fastener (For Non-Equipage Items).
MIL-R-3065	-	Rubber, Fabricated Parts.
MIL-C-3133	-	Cellular Elastomeric Materials, Fabricated Parts.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.
FED-STD-751	-	Stitches, Seams and Stitching.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-417	-	Rubber Compositions, Vulcanized, General Purpose, Solid, (Symbols and Tests).
MIL-STD-670	-	Classification System and Tests for Cellular Elastomeric Materials.

DRAWINGS

ARMY NATICK LABORATORIES

5-7-110	-	Case, Field Surveying Equipment, Steel; Cabinet, Storage, Assembly for Typewriter and Computing Machine.
5-7-111	-	Case, Field Surveying Equipment, Steel; Cabinet Assembly and Details for Azimuth Adding Machine.
5-7-112	-	Case, Field Surveying Equipment, Steel; Cabinet Assembly and Details for Calculating Machine.

NOTE: Miniature copies of above drawings, identified as figures 1 through 3 respectively, are attached for information purposes only.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.4).

3.2 Materials and components. Materials and components shall be as specified herein and on the drawings. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial cases provided the completed item complies with all provisions of this specification.

3.2.1 Steel. Steel used to fabricate all case panels and steel components not available commercially shall conform to 3.2.1.1 and 3.2.1.2.

3.2.1.1 Sheet and strip. Sheet and strip steel shall be hot rolled or cold rolled, conforming to QQ-S-698, with quality, temper or condition and finish optional.

3.2.1.2 Bars. Steel bars shall be either hot rolled or cold rolled conforming to QQ-S-630 or QQ-S-634, as applicable.

3.2.2 Threaded fasteners, washers and rivets. Threaded fasteners, washers and rivets shall be commercial items of the type, size, material and finish specified on the applicable drawings.

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3.2.3 Rubber. Rubber gasket material for type I and II cases shall conform to either of the following:

(a) Type, class, grade and suffix letter RS-315A of MIL-STD-417 and MIL-R-3065.

(b) Type, style and grade R015 of MIL-C-3133 and MIL-STD-670.

3.2.4 Adhesives, rubber. Adhesive used to cement rubber gaskets and crutch tips to metal shall be a solvent type, rubber adhesive conforming to type II of MMM-A-1617.

3.2.5 Webbing, cotton, 1 inch. Web straps for type III cases shall conform to type IIb, class 4, olive drab No. 7 color of MIL-W-530.

3.2.6 Clips, end, strap. End clips used for straps on type III cases shall conform to type I, class 1 or 2, size 1 inch of MIL-C-496.

3.2.7 Buckles, tongueless. Tongueless buckles used for web straps of type III cases shall conform to type II, style 1 or 4, class 1 or 3, size 1 inch of MIL-B-543.

3.2.8 Loops, strap fastener. Strap fastener loops for retaining web straps of type III cases shall conform to type V, size 1-1/4 inch of MIL-L-2045.

3.2.9 Thread, polyester. The thread used for all stitching shall conform to type I or II, class 1, size FF of V-T-285. The thread shall be dyed olive drab shade S-1 (C.A. 66022), and shall show good fastness to weathering equal to or better than the standard sample (see 6.3).

3.2.10 Tips, crutch. Crutch tips for type I cases shall be size 15, 3/8 inch I.D. of ZZ-T-351.

3.2.11 Wood. Wood for blocking on type I and II cases shall be free from decay, holes knots, larvae channels, shake, split, warp and brashness. The wood shall be any one of the following species:

Oak, red or white
Ash, white
Hickory
Beech
Birch, yellow
Maple

3.3 Design. The cases shall conform to the design, details and dimensions shown on the applicable drawings specified herein. Details not specifically covered shall be in accordance with the manufacturer's regular commercial practice

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3.3.1 Type I, case, field; adding machine. The adding machine case shall conform to Drawing 5-7-111. The location points for wood blocking (item 2-2) shall be furnished by the contracting officer, or located from the machine specified or furnished; or the blocking and screw fasteners shall be furnished unattached for future installation (see 6.2).

3.3.2 Type II, case, field; calculating machine. The calculating machine case shall conform to Drawing 5-7-112. Location for wood blocking shall be as specified in 3.3.1.

3.3.3 Type III, case, field; typewriter or computing machine. The typewriter and computing machine case shall conform to Drawing 5-7-110. The location point for the strap fastener loops (item 13), not dimensioned on the drawing, shall be furnished by the contracting officer or located from the machine specified or furnished; or the loops and rivets shall be furnished unattached for future installation (see 6.2).

3.4 Construction. Unless otherwise indicated on the drawings, all permanently attached metal components shall be joined by welds. Welds shall be of the types shown on the drawings and specified herein. Case corner construction not detailed on the drawings shall be optional with the manufacturer. All open corner joints shall be closed by continuous bead welds. When lap corner construction or metal angle binding construction or metal angle binding construction is furnished, spot welds, spaced not more than 3 inches center to center, may be utilized.

3.4.1 Stitching. Cotton webbing straps on type III cases shall be secured to strap fastener loops and buckles, as shown on the applicable drawing, by stitching with thread specified in 3.2.9. The stitching shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch. Thread breaks in stitching shall be overstitched not less than 1 inch at each break. Stitching shall be overstitched not less than 1 inch at the ends. If automatic machines are used, the box shall be finished with at least three (3) overlap or tie stitches. All thread ends shall be trimmed.

3.4.2 Rubber gaskets. Gaskets for type I and II cases shall be made from rubber specified in 3.2.3 and to the cross sectional dimensions shown on applicable drawings. The gasket shall completely fill the cover stop without stretching and shall be cemented in place, after finish enamel application is dry, using adhesive specified in 3.2.4, applied as recommended by the adhesives manufacturer.

3.4.3 Wood blocking. Installation of wood blocking for type I and II cases shall be accomplished after sealing of wood and enameling of cases has been completed.

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3.4.4 Draw pull catches and cover stop alignment. The draw pull catches and cover stops for type I and II cases shall be located so that newspaper shims cannot be withdrawn from between assembled base and cover when tested as specified in 4.5.1.

3.5 Finish. The end item shall be treated and painted in accordance with type A of MIL-T-704 except that finish color shall match semi-gloss color No. 24087, olive drab of FED-STD-595. All exposed rivet surfaces shall be coated with enamel after setting the rivet. The final applied finish shall be smooth, uniformly coated, free from scratches, sags, runs, blisters and foreign inclusions.

3.6 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Workmanship. The case shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, deteriorations or malformations. There shall be no sharp edges, burrs, slivers, or rough grinds. Gaskets shall completely fill the cover stop without stretching.

3.7.1 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds except spot welds shall be continuous. All welds shall be smooth and free from burn holes, cracks, undercuts and incomplete fusion. All scale and flux (when flux is used) shall be removed from the finished welds.

3.7.2 Fastening devices. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the field cases shall be examined for defects in table I, dimensions specified and tested in accordance with 4.5.1. The presence of any defect or failure of test shall be cause for rejection of first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1 above, component and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

a. Sealing of wood blocking (see 3.4.3).
b. Preparation of case for finish (see 3.5).

4.4.3 End item inspection. The inspection lot shall be all cases of one type offered for inspection at one time. The sample unit shall be one complete case.

4.4.3.1 Visual examination. The cases shall be examined for defects classified in table I. The inspection level shall be II and the AQL shall be 2.5 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish (organic coating)	Exposed rivet surfaces not enameled		X
	Scratch through to base metal		X
	Blistering, checking, or peeling		X
	Bare spots		X

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Finish (organic coating) (cont'd)	Tacky (not dry to touch)		X
	Wrong color		X
	Runs or sags		X
	Not smooth or uniform, foreign matter imbedded		X
Workmanship and construction	Any part missing	X	
	Component not attached as specified		X
	Cracks, splits, or splinters	X	
	Sharp edge or projection	X	
	Buckled, warped, malformed, or damaged	X	
Bolts, nuts and lockwashers	Thread stripped (type II cases)	X	
Welds	Not continuous, when applicable	X	
	Spot welds spaced on more than 3 inch centers		X
	Omitted where specified	X	
	Not smooth or free from pits, slag inclusions or porosity		X
Riveting (when applicable)	Mating surfaces not drawn tight		X
	Rivet missing	X	
	Heading poorly accomplished		X
	Splits		X
Identification marking	Missing, not permanent type, illegible, incomplete, incorrect, or in wrong location.		X

4.4.3.2 Dimensional examination. Inspection shall be made of the cases for compliance with dimensions specified. Any dimension not within tolerances specified shall be classified as a defect. The inspection level shall be S-2 and the AQL 4.0, expressed in terms of defects per hundred units.

4.4.3.3 End item testing. When a first article is not required, the first produced field case shall be tested as specified in 4.5.1. One sample field case from each production lot shall also be tested as specified in 4.5.1.

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Failure of the first produced item or the lot sample item shall result in rejection of all like manufactured items or rejection of the lot, as applicable.

4.4.4 Examination of preparation for delivery. Examination shall be made to determine that preservation, packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing, damaged, or otherwise defective
Workmanship	Inadequate application of components such as talcum powder; incomplete closure of container flaps, loose strapping, inadequate stapling; bulging or distortion of containers; sealing inadequate, improper or missing
Contents	Contents per container more or less than specified Contents of case not secured

4.5 Tests.

4.5.1 Cover fit, type I and II cases. Lay twenty-four (24) equally spaced strips of newspaper across the lip of the base. Newspaper strips shall be approximately 1 by 6 inch size. Latch the cover in place and attempt to withdraw each paper strip to determine compliance with 3.4.4. Withdrawal of any paper shims shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Types I and II. Each case shall have loose components secured so as to prevent movement while in transit. The rubber gasket of each case shall be uniformly covered with talcum powder to prevent adhesion during storage. The cover shall then be secured in position by means of the draw pull catch. Each case shall then be placed in a snug-fitting fiberboard box conforming to type CF or SF, class domestic, style RSC of PPP-B-636. Closure shall be in accordance with the box specification.

5.1.2 Level C. Each type I or II case shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Types I and II. Four cases of one type only, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to V3s style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification.

5.2.1.2 Type III. Each case shall have all the straps tightly buckled or fastened so as to prevent movement while in transit. Each case shall then be packed as specified in 5.2.1.1.

5.2.2 Level B.

5.2.2.1 Types I and II. Four cases of one type only, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF or SF, class domestic, style RSC of PPP-B-636. Each shipping container shall be closed in accordance with method II of PPP-B-636.

5.2.2.2 Type III. Each case shall have all the straps tightly buckled or fastened so as to prevent movement while in transit. Each case shall then be packed as specified in 5.2.2.1.

5.2.3 Level C. Cases shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The cases are intended to be used for storage of the various machines indicated in 1.2 in a mobile-control-section, surveying equipment, truck-type van.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) When a first article is required (see 3.1).
(d) Location points for wood blocking and strap fastener loops when applicable (see 3.3.1, 3.3.2 and 3.3.3).
(e) Applicable level of preservation and packaging and packing (see 5.1 and 5.2).

6.3 Shade samples. For access to standard shade samples of thread, address the procuring office issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodian:

Army - GL

Review activity:

Army - ME

User activity:

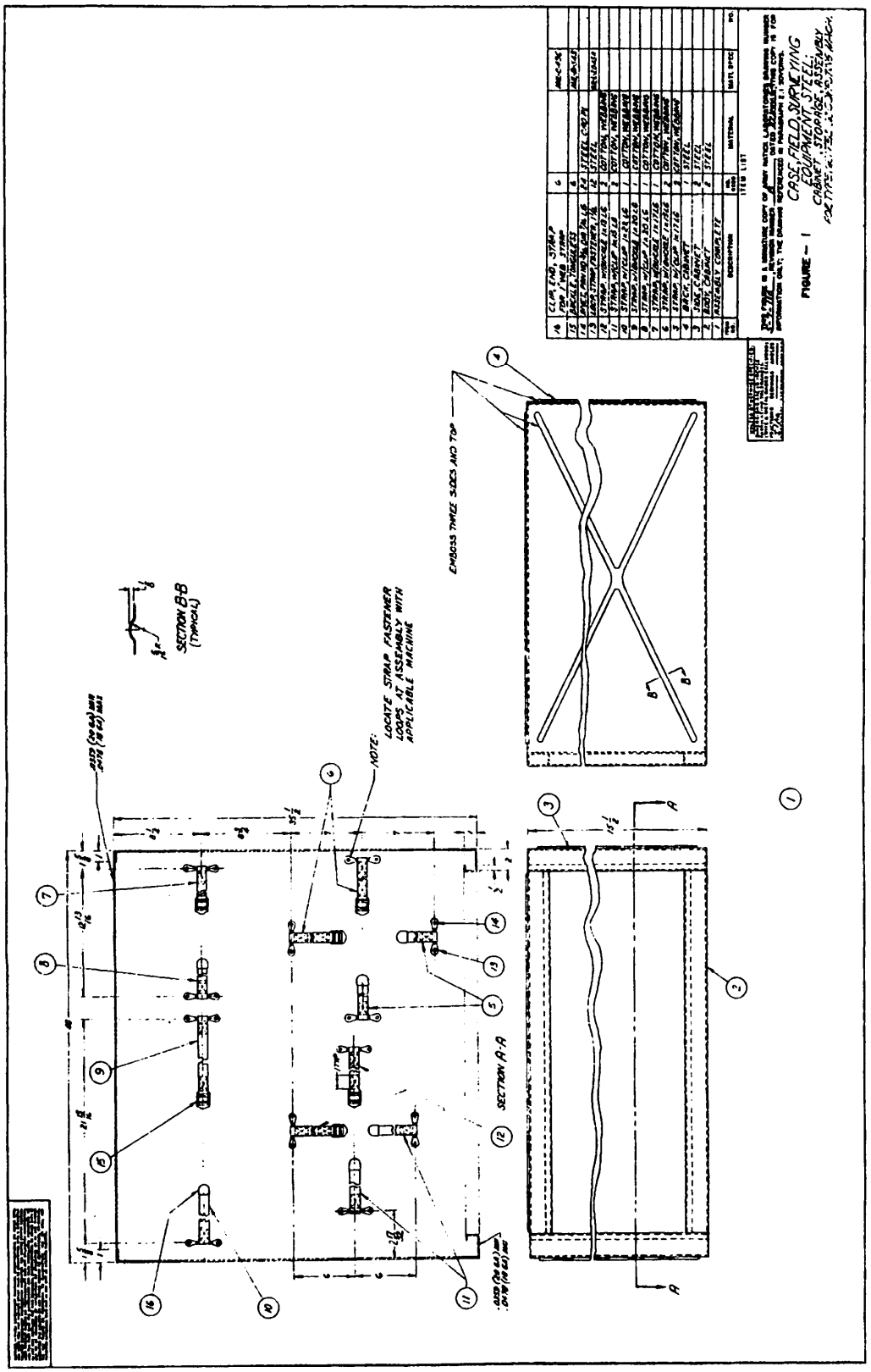
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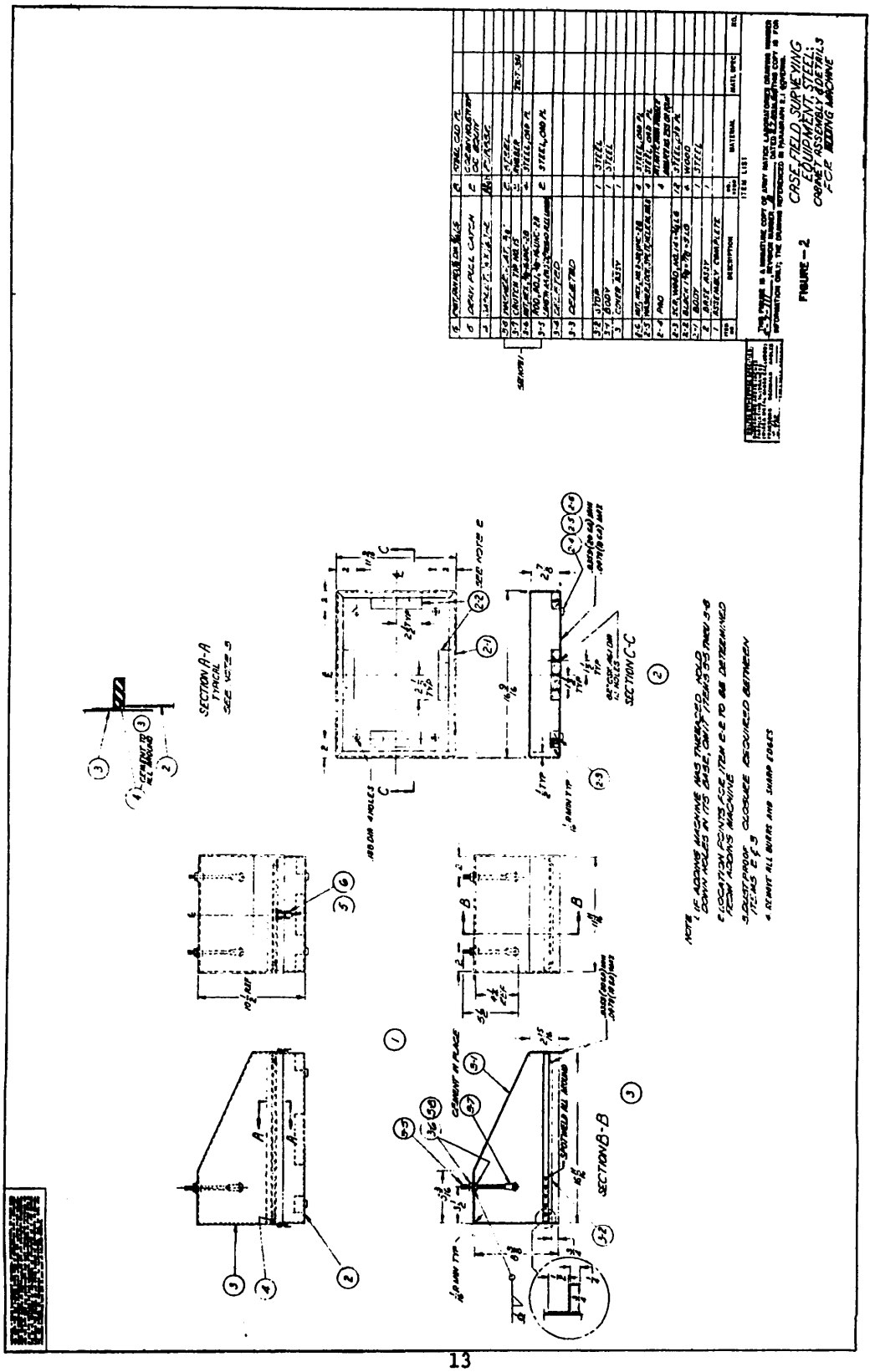


FIGURE - 2

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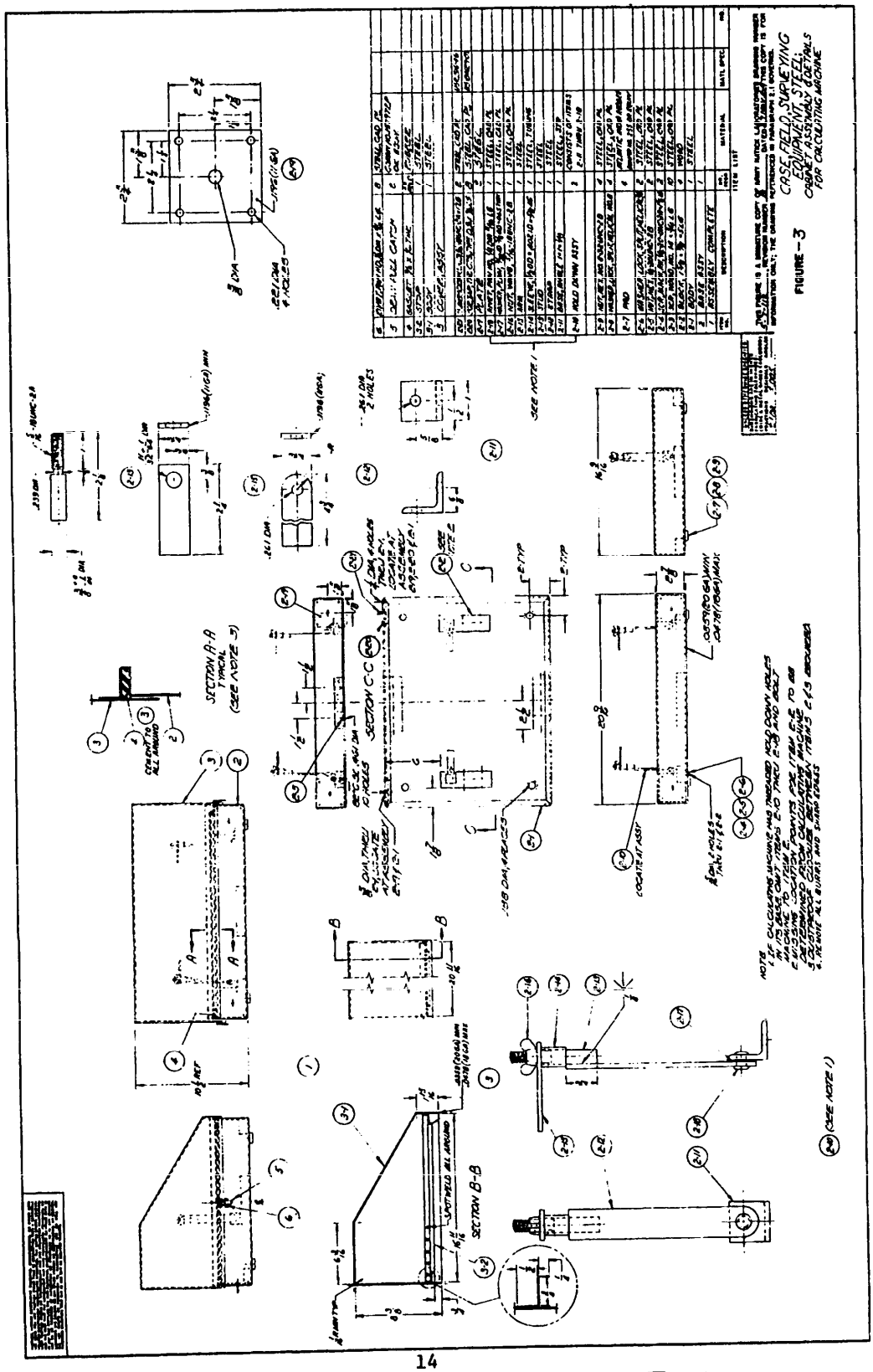


FIGURE -3
CASE, FIELD, SUPPLYING EQUIPMENT, STEEL: CABINET ASSEMBLY DETAILS FOR CALCULATING MACHINE

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CASES, FIELD, OFFICE MACHINES, SURVEYING EQUIPMENT, STEEL MIL-C-43293A(GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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